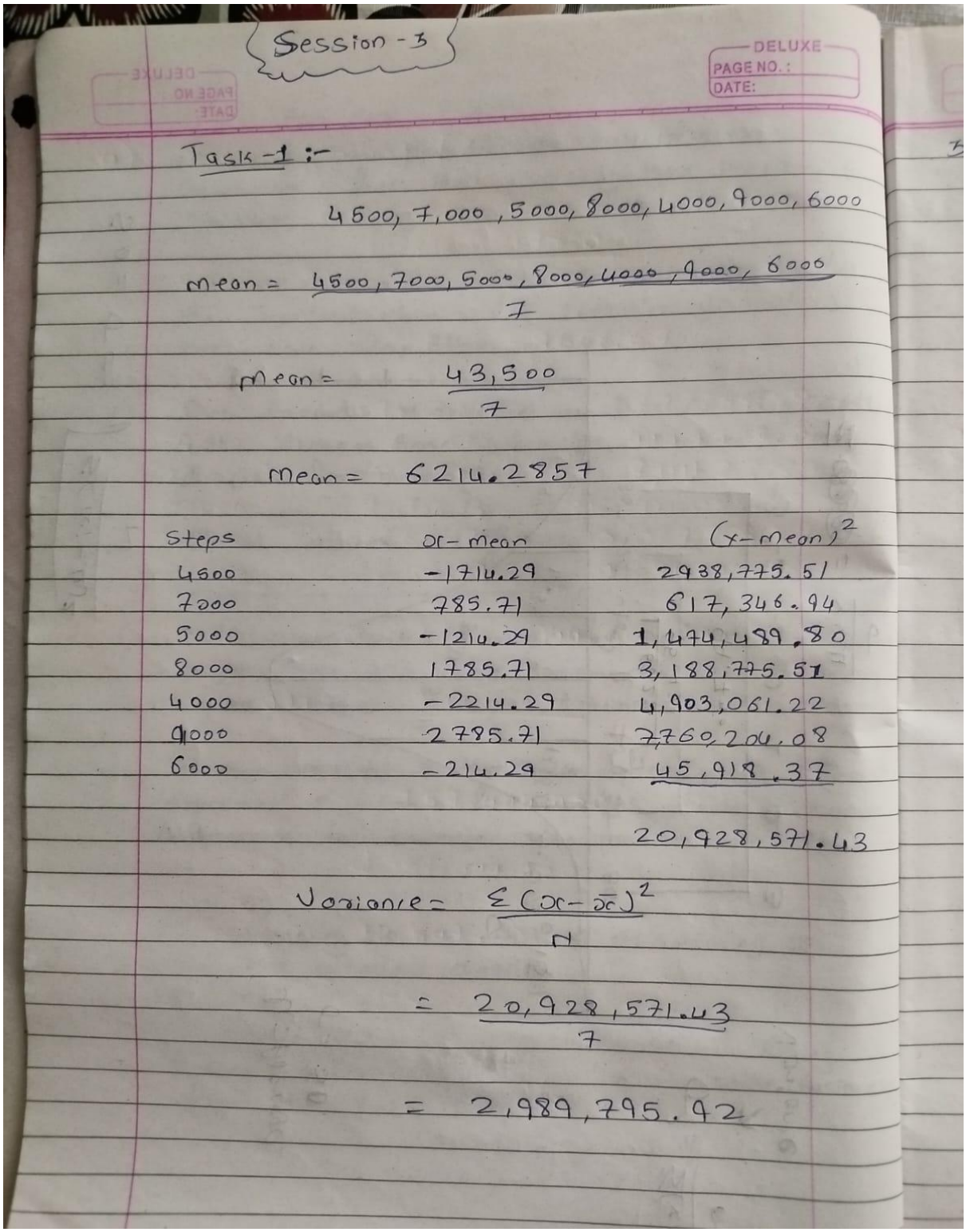
Session – 3

Task-1.

Given an array of daily step counts from your fitness app for one week (e.g., [4500, 7000, 5000, 8000, 4000, 9000, 6000]), calculate the variance by hand using the variance formula and show each calculation step.

Solution:



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Task-1 :-

4500, 7,000, 5000, 8000, 4000, 9000, 6000

$$\text{mean} = \frac{4500 + 7000 + 5000 + 8000 + 4000 + 9000 + 6000}{7}$$

$$\text{mean} = \frac{43,500}{7}$$

$$\text{Mean} = 6214.2857$$

Steps	x - mean	$(x - \text{mean})^2$
4500	-1714.29	2938,775.51
7000	785.71	617,346.94
5000	-1214.29	1,474,489.80
8000	1785.71	3,188,775.51
4000	-2214.29	4,903,061.22
9000	2785.71	7,760,204.08
6000	-214.29	45,918.37
		20,928,571.43

$$\text{Variance} = \frac{\Sigma (x - \bar{x})^2}{n}$$

$$= \frac{20,928,571.43}{7}$$

$$= 2,989,795.92$$

Task-2.

Write a Python function called calculate_standard_deviation(scores) that takes a list of exam scores and returns the standard deviation rounded to two decimal places.

Solution:

```
import statistics


def calculate_standard_deviation(scores):
    return round(statistics.stdev(scores), 2)


scores = [70, 80, 75, 90, 85]


result = calculate_standard_deviation(scores)


print("Standard Deviation:", result)
```

Output:

Standard Deviation: 7.91

Task-3.

Imagine two friends track their daily spending on Zomato over 5 days. Friend A spends [200, 200, 200, 200, 200] and Friend B spends [100, 300, 150, 400, 50]. Calculate the standard deviation for both and explain in 2-3 lines who has more consistent spending and why.

Solution:

3 friend A vs friend B

A = [200, 200, 200, 200, 200]

B = [100, 300, 150, 400, 50]

Mean = A = $\frac{1000}{5} = 200$

Variance = 0

SD = 0

B = Mean = $\frac{100+300+150+400+50}{5}$

$= 200$

Steps	$x - \bar{x}$	$(x-\bar{x})^2$
100	-100	10,000
300	100	10,000
150	-50	2,500
400	200	40,000
50	-150	22,500
		85,000

Var = $\frac{85,000}{5} = 17,000$

SD = $\sqrt{17,000}$

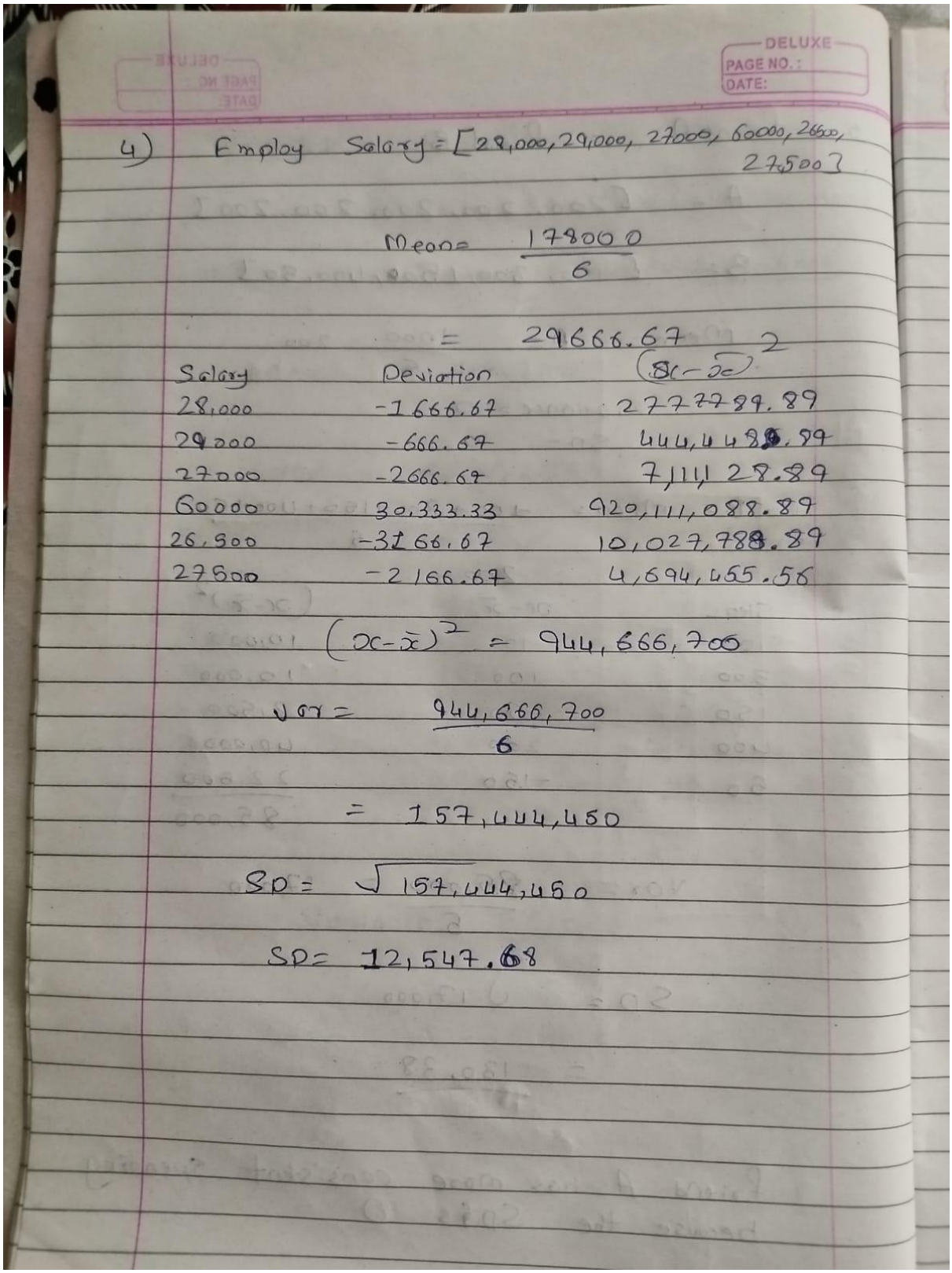
$= 130.38$

friend A has more consistent spending because the SD is 0

Task-4.

You are given the monthly salary data of 6 employees in a startup: [28000, 29000, 27000, 60000, 26500, 27500]. Calculate the variance and standard deviation, then interpret in 2-3 lines what the high or low spread means for this team.

Solution:



4) Employ Salary = [28,000, 29,000, 27000, 60000, 26500, 27,500]

$$\text{Mean} = \frac{178000}{6}$$

$$= 29666.67$$

Salary	Deviation	$(x-\bar{x})^2$
28,000	-1666.67	2777789.89
29000	-666.67	444,448.89
27000	-2666.67	7,111,128.89
60000	30,333.33	920,111,088.89
26,500	-3166.67	10,027,788.89
27500	-2166.67	4,694,455.56

$$(x-\bar{x})^2 = 944,666,700$$

$$\text{Var} = \frac{944,666,700}{6}$$

$$= 157,444,450$$

$$SD = \sqrt{157,444,450}$$

$$SD = 12,547.68$$